Prediction of Chronic Disease Risk Using Nutrition, Lifestyle & Clinical Data from The Irish Longitudinal Study on Ageing (TILDA)

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Abstract

Falls, frailty, and chronic conditions such as cardiovascular disease, diabetes, and obesity remain leading causes of morbidity and mortality in older adults. Yet, existing predictive models often rely narrowly on physical health indicators, overlooking nutrition and lifestyle factors that may enhance early detection of risk. This thesis develops and evaluates machine learning models using data from The Irish Longitudinal Study on Ageing (TILDA) to predict key health outcomes in a cohort of older adults. The models integrate physical, mental, lifestyle, and socio-demographic predictors, with particular emphasis on under-explored nutritional and behavioural variables highlighted in prior research. Multiple approaches are compared, including logistic regression, random forests, and gradient boosting, with performance assessed via cross-validation and feature importance analysis. By identifying high-impact, modifiable predictors, this work aims to advance early risk stratification, guide targeted prevention strategies, and inform public health policy to support healthy ageing.

Declaration

I Abderahman Haouit declare that I have produced this thesis without the prohibited assistance of third parties and without making use of aids other than those specified; notions taken over directly or indirectly from other sources have been identified as such. This thesis has not previously been presented in identical or similar form to any other Irish or foreign examination board.

The thesis work was conducted from 2025 to 2026 under the supervision of Dr. Alison O'Connor at University of Limerick.

Limerick, 2026

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Your dedication goes here. Open the dedication.tex file found at:
0_frontmatter/dedication.tex

Contents

List of Tables

LIST OF TABLES

List of Figures

List of Abbreviations

AI	Artificial Intelligence
AUC	Area Under the ROC Curve
CES-D	Center for Epidemiologic Studies Depression Scale
DNN	Deep Neural Network
ELSA	The English Longitudinal Study of Aging
EU	European Union
GBM	Gradient Boosting Machine
GDPR	General Data Protection Regulation
HADS	Hospital Anxiety and Depression Scale
HRS	The Health and Retirement Study, (a longitudinal study of ageing in the United States)
IDS	The Intellectual Disability Supplement
ISSDA	The Irish Social Science Data Archive
LDA	Linear Discriminant Analysis
ML	Machine Learning
RF	Random Forest
SHAP	Shapley Additive Explanations
SHARE	The Survey of Health, Aging and Retirement in Europe
TILDA	The Irish Longitudinal Study on Ageing

List of Abbreviations

WHO World Health Organization

Chapter 1

Introduction

1.1 Background and Context

Chronic conditions-including cardiovascular disease, diabetes, and frailty—are leading contributors to morbidity and mortality in ageing populations [1, 2]. These conditions are associated with declines in functional capacity (e.g., gait speed, grip strength [3]), increased healthcare utilisation [4], and reduced quality of life [5].

The financial and societal burden is substantial. In Europe, poor diet is a major risk factor for non-communicable diseases, contributing billions in healthcare costs annually [6]. In Ireland, malnutrition alone is estimated to cost over €1.4 billion per year [7]. Recognising this, international and national policies—including the World Health Organization (WHO) Global Action Plan on Noncommunicable Diseases and Ireland's *Healthy Ireland* framework—highlights nutrition and lifestyle interventions as central to healthcare strategy.

In parallel, (Artificial Intelligence (AI)) and (Machine Learning (ML)) are increasingly applied in healthcare to support risk prediction and personalised intervention. However, European regulations such as the General Data Protection Regulation (GDPR) and the forthcoming European Union (EU) AI Act impose strict requirements for transparency, explainability, and governance. This regulatory context underscores the importance of interpretable and ethically robust ML approaches.

Traditional risk prediction models have largely relied on clinical biomarkers such as blood pressure and lipid profiles. Yet, growing evidence shows that broader predictors—including nutrition [8], physical activity [9], mental health [10], and social determinants [11]—enhance predictive performance.

(The Irish Longitudinal Study on Ageing (TILDA))[1] provides a nationally representative cohort of adults aged 50+ in Ireland, surveyed across multiple domains in repeated waves (waves 1–5). The raw wave datasets capture the full richness of TILDA-specific measures relevant to Irish health, policy, and society. In parallel, the Harmonized TILDA dataset restructures selected variables to align with the RAND (The Health and Retirement Study, (a longitudinal study of ageing in the United States) (HRS)) framework, enabling cross-national comparison with equivalent studies such as HRS (US), The English Longitudinal Study of Aging (ELSA) (UK), The Survey of Health, Aging and Retirement in Europe (SHARE) (continental Europe). This harmonisation enhances international comparability, though at the cost of excluding some study-specific variables. An additional dataset, the The Intellectual Disability Supplement (IDS), extends coverage to individuals with disabilities, improving inclusivity and reducing sampling bias. Together, these resources offer both detailed national data and a platform for internationally comparable, generalisable analyses.

Despite these advantages, existing ML research on chronic disease prediction faces persistent limitations: over-reliance on clinical predictors, limited incorporation of lifestyle and social variables, predominantly cross-sectional study designs that fail to capture trajectories, and a lack of interpretability consistent with emerging AI governance standards.

This thesis addresses these gaps by integrating TILDA's longitudinal, multidomain data with interpretable ML methods, demonstrating how nutrition, physical activity, and mental health can be combined with traditional clinical markers to improve prediction of ageing-related outcomes.

[1]TILDA data are pseudonymised and accessed under licence via The Irish Social Science Data Archive (ISSDA). Use adheres to UL S&E ethics approval and data governance requirements.

1.2 Research Problem and Motivation

Despite increasing policy interest in lifestyle and nutrition and advances in ML methods, approaches to chronic disease prediction in older adults remain constrained in several ways.

1.2.1 Limited Integration of Multidimensional Data

Most models prioritise clinical variables while underutilising modifiable lifestyle and nutrition factors [8, 12].

1.2.2 Underuse of Longitudinal Dynamics

Few studies exploit repeated measures to capture trajectories of risk, such as frailty progression [13].

1.2.3 Lack of Explainability

Many ML models are not interpretable, limiting clinical adoption and failing to meet emerging regulatory requirements for trustworthy AI in healthcare [14].

Addressing these challenges is essential to improve predictive accuracy, support evidence-based interventions, and ensure alignment with evolving policy and regulatory priorities.

1.3 Research Aims and Objectives

This thesis develops and validates ML models that integrate nutrition, lifestyle, and mental health features with clinical data to improve prediction of chronic disease and ageing-related outcomes in a representative cohort of older adults.

- Review existing applications of ML for chronic disease prediction in ageing populations, with emphasis on gaps in lifestyle and nutrition integration.

- Preprocess and harmonise TILDA waves 1-5, and engineer multidomain features (nutrition, activity, mobility, mental health, clinical indicators).

- Develop and compare baseline statistical models (e.g., logistic regression) with advanced ML methods (Random Forest (RF), Gradient Boosting Machine (GBM)).
- Evaluate models using discrimination (Area Under the ROC Curve (AUC)), calibration, and error metrics (accuracy, precision, recall, F1-score), perform ablation analyses to assess the incremental value of lifestyle and nutrition predictors.
- Identify actionable predictors (e.g., vitamin D status, gait speed reserve) and highlight modifiable risk factors for intervention.
- Document ethical, governance, and reproducibility practices for handling longitudinal pseudonymised health data.

1.4 Contributions of the Research

This research makes contributions at three levels:

- **Empirical:** Demonstrates how nutrition, lifestyle, and mental health features enhance prediction when combined with established clinical baselines.
- **Methodological:** Provides a reproducible pipeline for harmonising longitudinal TILDA data and integrating multidomain features into ML workflows.
- **Clinical:** Identifies modifiable predictors with relevance for targeted interventions and policy to promote healthy ageing.

1.5 Thesis Outline

- *Chapter Two:* Literature review on ML prediction of chronic disease in ageing populations, with emphasis on lifestyle and nutrition predictors and methodological considerations.

- *Chapter Three:* TILDA study design, data preprocessing, harmonisation, and feature engineering.(Flow Charts..)
- *Chapter Four:* Modelling framework, including baseline and advanced ML, hyperparameter tuning, validation, and evaluation metrics.(Code snippets..)
- *Chapter Five:* Results, model comparison, feature importance, and ablation analyses.
- *Chapter Six:* Discussion of implications, limitations, ethical considerations, and future directions.

Supporting documents are included in the appendix:

- *Appendix A*: Variable codebook and wave harmonisation notes (including anonymisation actions, top-coding, and grouping rules relevant to the features used).
- *Appendix B*: Ethics.
- *Appendix C*: Modelling details, supplementary tables/figures, and reproducibility checklist (overview of data processing scripts).

A digital archive accompanies this dissertation (access request required):

- *Folder 1*: Preprocessing and feature-engineering scripts (documentation only, no raw TILDA data).
- *Folder 2*: Model training/evaluation code and configuration files to reproduce reported results.

Chapter 2

Literature Review

2.1 Introduction

This chapter reviews literature on ageing, chronic disease, and predictive modelling using TILDA. The focus is on three domains central to later-life health: frailty and physical function, mental health, and nutrition and biological ageing. The review also considers how machine learning has been applied to these areas and identifies gaps that motivate the present study.

2.2 Chronic Disease and Ageing

Chronic conditions are the main drivers of morbidity and reduced quality of life in older adults. Multimorbidity, frailty, and mental health difficulties often coexist, compounding risks of functional decline. In line with the WHO healthy ageing framework, research using TILDA highlights frailty, mental health, and nutrition as interconnected determinants of later-life outcomes. The following subsections examine each domain in turn.

2.2.1 Frailty and Physical Function

Frailty reflects reduced physiological reserve and heightened vulnerability to adverse outcomes. In TILDA, it has been measured using both the frailty phenotype and the frailty index, with gait speed, grip strength, and mobility decline as key indicators. These markers are attractive because they are easy to collect at scale and strongly predict morbidity and mortality.

Longitudinal analyses show that frailty is not a fixed trajectory but a dynamic process. Multi-state modelling, for instance, revealed that many prefrail individuals transition back to a non-frail state, while frailty itself is associated with high mortality risk [15]. Other work links metabolic syndrome to increased risk of frailty, underscoring the influence of cardiometabolic health [16]. At the same time, predictors such as obesity and vision problems show age-dependent effects, with obesity exerting stronger influence in midlife and protective factors like grip strength becoming more important later in life [17]. Despite these insights, the evidence base is constrained by observational designs, reliance on self-report, and the tendency to analyse predictors in isolation. External validation beyond Ireland is also limited, which raises questions of generalisability. These issues highlight the need for integrative approaches that capture the multidimensionality of frailty and point to the potential role of machine learning methods (Chapter 4) to model frailty transitions with greater precision.

2.2.2 Mental Health in Older Adults

Mental health is a critical determinant of quality of life in older adults, with depression, anxiety, loneliness, and social isolation commonly observed in this population. Evidence from TILDA demonstrates that mental health outcomes are intricately linked to both physical health and social engagement. For instance, studies leveraging Wave 1-2 data indicate that higher engagement in physical, social, and cognitive activities—captured via composite ABC indicators—was associated with lower prevalence and incidence of depression and anxiety [18]. This suggests that lifestyle and social integration act as protective factors against mental health decline.

However, there are nuances in these findings. While social participation often buffers against anxiety and depressive symptoms, the effect on cognitive impairment appears weaker and less consistent, highlighting domain-specific variability. Moreover, some studies report that self-reported measures of mental health and activity may introduce bias, potentially inflating associations [18], [19]. These limitations underscore the need for objective and multi-modal assessment methods.

Physical activity emerges as a particularly robust predictor. Analyses of Wave 1-5 datasets reveal that higher accelerometer-measured moderate-to-vigorous physical activity correlates with lower depressive symptoms, even when controlling for demographic and clinical covariates [9]. Notably, these associations appear dose-dependent, indicating that even sub-recommended activity levels can confer measurable mental health benefits. This aligns with findings from longitudinal cohorts showing that incident falls and functional decline predict subsequent depressive symptoms, mediated partly by fear of falling [20]. These studies illustrate the complex interplay between mental health, physical function, and risk exposure in ageing populations.

Critically, while these findings demonstrate clear associations, the literature is dominated by observational designs with limited external validation. The majority of studies focus on adults aged 50+, predominantly within the Irish context, which limits generalisability to more ethnically and socioeconomically diverse populations. Additionally, mental health constructs are frequently operationalised via broad self-report scales Center for Epidemiologic Studies Depression Scale (CES-D), Hospital Anxiety and Depression Scale (HADS), without corroborating clinical or biomarker data. Taken together, these gaps highlight the importance of integrating objective, longitudinal measures of mental health alongside multidimensional predictors to improve the predictive accuracy and interpretability of models (see Chapter 4 for ML approaches).

2.2.3 Nutrition and Biological Ageing

Nutrition is increasingly recognised as a modifiable determinant of biological ageing and frailty. Within TILDA, key nutritional biomarkers—such as vitamin D, lutein, and zeaxanthin—have been linked to functional and cognitive outcomes [8], [21], [22]. For example, plasma lutein and zeaxanthin levels measured in Wave 1-5 were associated with slower frailty progression and improved physical performance, suggesting that blood levels of these carotenoids reflect long-term exposure to a healthy-promoting diet. [8]. Similarly, low vitamin D status was consistently related to increased risk of type 2 diabetes and musculoskeletal deficits, particularly among older adults with limited sunlight exposure [21], [22].

Malnutrition, operationalised via BMI thresholds and unintentional weight loss, also predicts adverse outcomes. Evidence indicates that functional decline, hospitalisation, and limited social support increase malnutrition risk, with differential effects by sex [2]. These findings suggest that interventions targeting both dietary intake and social determinants may mitigate nutritional risk, yet the literature rarely integrates these variables comprehensively.

The distinction between chronological and biological ageing is central to understanding these nutritional impacts. Biological age, estimated using biomarkers such as epigenetic clocks or frailty indices, often diverges from chronological age, reflecting cumulative environmental, lifestyle, and physiological exposures [23], [24]. In this context, metabolic syndrome and nutritional deficiencies have been linked to accelerated biological ageing, highlighting potential intervention targets to slow physiological decline [23], [14]. Notably, these studies emphasise that biological ageing measures provide finer-grained predictions of vulnerability than chronological age alone, particularly when integrated with multi-domain predictors in longitudinal cohorts.

Critically evaluating the literature reveals several limitations. Most studies are cross-sectional, with limited temporal resolution to capture diet-induced changes over time. Sample sizes for nutritional biomarker studies are often smaller than those for functional assessments, reducing statistical power. Additionally, while the relationships between nutrition and ageing outcomes are generally consistent, effect sizes vary, and causality remains difficult to establish due to confounding by comorbidities, socioeconomic factors, and lifestyle behaviours. These gaps point to opportunities for more sophisticated, integrative modelling approaches, including machine learning pipelines that combine nutritional biomarkers with functional, clinical, and psychosocial data.

2.3 Machine Learning in Ageing and Chronic Disease

- Overview: Why ML is valuable in ageing research [25]
- Transition from regression-based models → ensemble ML → deep learning(cite a example here)

2.3.1 Feature Sets in ML Studies

- Clinical, sociodemographic, lifestyle, cognition, psychosocial
- Over-reliance on physical health predictors vs underuse of nutrition/social variables

2.3.2 ML Models Used

- Logistic regression, Markov models, Linear Discriminant Analysis (LDA) (traditional)
- Ensembles RF, GBM, Deep Neural Network (DNN) – e.g., ([26]) diabetes prediction
- Unsupervised clustering for dementia ([27])
- Brain age prediction models ([28], [29])

2.3.3 Performance and Validation

- AUC, R^2 ranges across tasks (diabetes, gait speed, mortality, depression)
- Limitations: internal validation only, cross-sectional data, no generalisation

2.3.4 Explainability in ML for Ageing

- Shapley Additive Explanations (SHAP), TreeExplainer examples ([26], [30], [3])
- Challenges of black-box models in clinical practice

2.4 Gaps in the Literature

- Limited population diversity (mostly Irish, 50+)
- Over-reliance on physical predictors; neglect of nutrition, social, religious/spiritual, environmental variables
- Dominance of regression models; limited advanced ML such as neural networks, transformers
- Validation gaps – very few external validations, especially for ensemble and brain-age models
- Underexplored outcomes: pain, sleep, QoL
- Cross-sectional bias; weak time-series modelling
- Poor transparency on missing data handling
- Limited integration of nutrition biomarkers
- Lack of biological age prediction vs chronological
- Interpretability challenges in ML

2.5 Chapter Summary

- I need to recap the main findings from LR systematic mapping excel file
- Identify your contribution niche:
  - integrating nutrition + lifestyle + clinical predictors
  - ML + explainability
  - time-based prediction (longitudinal TILDA data)
- Lead into Methodology (Chapter 3)

Chapter 3

Data and Preprocessing

3.1 Insert a Figure

We refer to a figure in the text like this: (Figure 3.1).

Figure 3.1: Title of Figure - Description. [Source: (**?**)]

3.2 Creating a list

- **(2000)** item 1.
- **(2004)** item 2.
- **(2010)** item 3.
- **(2013)** item 4.

3.2.1 Creating a Table

Table 3.1: Table Title

	Resolution	Min	Max
Gyroscope	4.5mV / °/s	0.27 °/s	406 °/s
Accelerometer	600mV /g	0.002g	2g
Magnetometer	385mV/ gauss	0.317 gauss	6 gauss

3.3 Quote

An in-text quote is declared in the following way:

> Recycling is the way forward! Recycling is the way forward! Recycling is the way forward! Recycling is the way forward! Recycling is the way forward! Recycling is the way forward! Recycling is the way forward! Recycling is the way forward! Recycling is the way forward! Recycling is the way forward! Recycling is the way forward! Recycling is the way forward! Recycling is the way forward! Recycling is the way forward! Recycling is the way forward! Recycling is the way forward! Recycling is the way forward! Recycling is the way forward! Recycling is the way forward! (CommonSense, p. 0)

Another way of doing it is:

There are in our existence spots of time,
That with distinct pre-eminence retain
A renovating virtue, whence-depressed
By false opinion and contentious thought,
Or aught of heavier or more deadly weight,
In trivial occupations, and the round
Of ordinary intercourse-our minds
Are nourished and invisibly repaired;
A virtue, by which pleasure is enhanced,
That penetrates, enables us to mount,
When high, more high, and lifts us up when fallen.
(**?** , verses 208-218)

Chapter 4

Methodology

4.1 Math

In-text Math

- A vector vector $\hat{A}_{ba}\{x_{ba}, y_{ba}, z_{ba}\}$
- Another vector $R^b\{\hat{t}_{1b}, \hat{t}_{2b}, \hat{t}_{3b}\}$ as:

A series of Equations:

$$\hat{t}_{1b} = \hat{A}_{ba} \tag{4.1}$$

$$\hat{t}_{2b} = \frac{\hat{A}_{ba} \times \hat{M}_{bm}}{\mid \hat{A}_{ba} \times \hat{M}_{bm} \mid} \tag{4.2}$$

$$\hat{t}_{3b} = \hat{t}_{1b} \times \hat{t}_{2b} \tag{4.3}$$

$$R^a = R^b(R^i)^T = [\hat{t}_{1b}, \hat{t}_{2b}, \hat{t}_{3b}][\hat{t}_{1i}, \hat{t}_{2i}, \hat{t}_{3i}]^T \tag{4.4}$$

where the symbol ^{T} denotes transposition.

$$\hat{t}_{2b} = \frac{\hat{A}_{ba} \times \hat{M}_{bm}}{\mid \hat{A}_{ba} \times \hat{M}_{bm} \mid} \tag{4.5}$$

$$= \frac{(y_{ba}z_{bm} - y_{bm}z_{ba}, z_{ba}x_{bm} - z_{bm}x_{ba}, x_{ba}y_{bm} - x_{bm}y_{ba})}{\mid A_{ba} \mid\mid M_{bm} \mid \sqrt{1 - (\hat{A}_{ba} \cdot \hat{M}_{bm})^2}}$$

$$= \frac{-0.2338, -0.0931, -0.0651}{0.2601}$$

$$= (-0.8989, -0.3579, -0.2503)$$

which if normalised gives:

$$= (-0.8995, -0.3581, -0.2504)$$

A matrix

$$R^b = [\hat{t}_{1b}, \hat{t}_{2b}, \hat{t}_{3b}] = \begin{pmatrix} -0.2698 & -0.8995 & 0.3439 \\ 0.0035 & -0.3581 & -0.9337 \\ 0.9629 & -0.2504 & 0.0997 \end{pmatrix}$$

Chapter 5

Results

5.1 References

Open the file 'references_myphd' with BibTex. The file is located in '/7_references/'

The file contains a series of templates for the proper formatting of references according to the Harvard referencing style. Click on each reference in the text below to see how Latex has formatted the reference on the References section.

- **Book**: (**?** , p. 10) (page number required only if referring to an in-text quote)
- **Book Chapter**: (**?**)
- **Book Contribution**: (**?**)
- **MUSIC CD/DVD**: (**?**)
- **Journal Article**: (**?**)
- **Patent**: (**?**)
- **PhD/Master Dissertation**: (**?**)
- **Conference Paper**: (**?**)
- **Technical Report**: (**?**)

- **Unpublished Report**: (**?**)
- **Web Videos**(**?**) (. . . not the name of the person who uploaded the video though)
- **Web Paper**: (**?**)
- **Web Journal Article**: (**?**)
- **Webpage**: (**?**)
- **Downloaded Images or Sounds**: (**?**)

You can also you this kind of referencing if needed in the text:
Surname [**?**] said that . . .

Chapter 6

Conclusions and Future Directions

6.1 Summary

6.2 Conclusions

6.3 Contributions

6.4 Future Work

Appendix A

Insert a figure

Appendix B

Title

Type here you text as usual . . .

Appendix

Appendix C

Code for estimating attitude

AHRS_TRIAD.C

```
/*
 *
 *  Copyright (c) 2013, Giuseppe Torre
 *  All rights reserved.
 *
 *  Redistribution and use in source and binary forms, with or without
 *  modification, are permitted provided that the following conditions are met:
 *      * Redistributions of source code must retain the above copyright
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 *  DIRECT, INDIRECT, INCIDENTAL, SPECIAL, EXEMPLARY, OR CONSEQUENTIAL DAMAGES
 *  (INCLUDING, BUT NOT LIMITED TO, PROCUREMENT OF SUBSTITUTE GOODS OR SERVICES;
 *  LOSS OF USE, DATA, OR PROFITS; OR BUSINESS INTERRUPTION) HOWEVER CAUSED AND
 *  ON ANY THEORY OF LIABILITY, WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT
 *  (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE OF THIS
 *  SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.
 *
 */



#include "ext.h"
#include "ext_mess.h"
#include <string.h>
#include <stdio.h>
#include <math.h>
#define EPSILON 1e-6

//******************CALIBRATION STUFF*************************//
/* INTRODUCE THE OFFSET VALUES IN THE FOLLOWING VARIABLES IN ADC values */
#define ACCELX_OFFSET 2043 //initial offset in accelerometer X
#define ACCELY_OFFSET 2053 // initial offset in accelerometer Y
#define ACCELZ_OFFSET 2264//initial offset in accelerometer Z

#define MagX_OFFSET 1917 //initial offset in Magnetometer X
```

Appendix

```
#define MagY_OFFSET 1813 // initial offset in Magnetometer Y
#define MagZ_OFFSET 1667//initial offset in Magnetometer Z
/* WE SHOULD CALIBRATE THE FOLLOWING VALUES FOR EACH PARTICULAR IMU axis */
#define ACCELX_Resolution 536
#define ACCELY_Resolution 661
#define ACCELZ_Resolution 559
#define MagX_Resolution 186
#define MagY_Resolution 189
#define MagZ_Resolution 170
//****************** END CALIBRATION STUFF  ************************//


void *this_class; // Required. Global pointing to this class

typedef struct _triad // Data structure for this object
{
        t_object m_ob;   // Must always be the first field; used by Max

        Atom m_args[9];  // we want our inlet to be receiving a list of 10 elements

        long m_value;    //inlet

        void *m_R1;            //these are all the outlets for the 3 X 3 Matrix
        void *m_R2;            // R1 --> firts top left cell ..R2 middle cell of the first
            row ...and so on
        void *m_R3;                                    //
        void *m_R4;                                    // End Outlets
            ______________________________________________
} t_triad;

                        void *triad_new(long value);


                        void triad_assist(t_triad *triad, void *b, long msg, long arg, char *
                            s);
                        void triad_free(t_triad *triad);



                        void triad_list(t_triad *x, Symbol *s, short argc, t_atom *argv);




                        void MatrixByMatrix(double *Result,double *MatrixLeft,double *
                            MatrixRight);

                        void Matrix2Quat(double *Quat,double *Matrix);
                        void Quat2Matrix(double *Matrix, double *Quat);
                        void inverseQuat(double *InvQuat, double *RegQuat);
                        void NormQuat(double *YesQuat, double *NotQuat);
                        void Slerp(double *NewQuat, double *OldQuat, double *CurrentQuat);
                        void NormVect(double *YesVect, double *NotVect);
                        double orientationMatrix[9];
                        double Result[9];
                        int     i;
                        double InvorientationMatrix[9];

                        double temp[6];
                        double ref[6];
                        double vectAx[3];
                        double vectAy[3];
                        double vectAz[3];
                        double vectBx[3];
                        double vectBy[3];
                        double vectBz[3];
                        double MagnCrosProd_A;
                        double MagnCrosProd_B;
                        double accnorm, magnorm, earthnorm, VectAynorm, VectAznorm,
                            VectBynorm, VectBznorm;
```

```
                            double m[9], n[9];
                            double quat_e[4];
                            double invquat_e[4];
                            double mult;
                            double quat_new[4];
                            double quat_old[4];
                            double ecs,accex_ADCnumber,y,accey_ADCnumber,z,accez_ADCnumber,mx,
                                 magnx_ADCnumber,my,magny_ADCnumber,mz,magnz_ADCnumber;
// SLERP Variables
                            double trace, Suca;
                            double tol[4], omega, sinom, cosom, scale0, scale1, tez,
                                 orientationMatrixA[9];

int main(void)
{
         // set up our class: create a class definition
         setup((t_messlist**) &this_class, (method)triad_new, (method)triad_free, (short)
              sizeof(t_triad), 0L,A_GIMME, 0);
    addmess((method)triad_list, "list", A_GIMME, 0);
         addmess((method)triad_assist, "assist", A_CANT, 0);
         finder_addclass("Maths","triad");
         post("....I'm-TRIAD-Object!....from-AHRS-Library...",0);
         return 0;
}

/* --------------triad_new ----------------*/

void *triad_new(long value)
{
         t_triad *triad;
         triad = (t_triad *)newobject(this_class);        // create the new instance and return
               a pointer to it
         triad->m_R4 = floatout(triad);
         triad->m_R3 = floatout(triad);
         triad->m_R2 = floatout(triad);
         triad->m_R1 = floatout(triad);

return(triad);
}

etc.

etc.
```

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